Regularized Linear Prediction of Speech

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***Abstract*—All-pole spectral envelope estimates based on linear prediction (LP) for speech signals often exhibit unnaturally sharp peaks, especially for high-pitch speakers. In this paper, regularization is used to penalize rapid changes in the spectral envelope, which improves the spectral envelope estimate. Based on extensive experimental evidence, we conclude that regularized linear prediction outperforms bandwidth-expanded linear prediction. The regularization approach gives lower spectral distortion on average, and fewer outliers, while maintaining a very low computational complexity.**

***Index Terms*—Bandwidth expansion, envelope estimation, linear prediction (LP), regularization.**

I. Introduction

LINEAR prediction (LP) is commonly used to estimate the parameters of an autoregressive (AR) model describing the spectral envelope [1]. A well-known problem with the LP approach is that the resulting spectral envelope can have very sharp peaks for speakers with high pitch frequency [2]. The envelope estimation fails to separate the short-term dependency (the envelope) from the long-term dependency (the pitch), and the resulting envelope is contaminated with the harmonics. These peaky spectral envelopes can cause problems in speech modification (e.g., pitch scaling [3]), speaker recognition [4], [5], and speech coding [6].

The problem with nonsmooth spectral envelopes is usually avoided by one of two approaches. The first is to use a different spectral estimation method, not based on LP (e.g., [2], [7]). The second, which we will refer to as conventional LP methods, is to use a heuristic modification to LP. These methods include 1) using a low-order AR model, 2) lag windowing of the autocorrelation sequence and/or 3) bandwidth expansion by scaling the radii of the poles of the AR model. None of these methods can claim to be optimal in any sense, and they do not solve the problem of harmonic contamination.

Lag windowing of the autocorrelation sequence [8], [9] consists of multiplication of the autocorrelation sequence $r(m)$ by a lag window $w_{\text{lag}}(m)$ to obtain modified autocorrelation values $r'(m) = w_{\text{lag}}(m)r(m)$. This approach is commonly used in speech coders, e.g., ITU-T G.729 [10], 3GPP adaptive multirate (AMR) narrow-band [11], and 3GPP AMR wide-band [12], which all use an exponential window w_{lag} that yields $f_0 = 60$ Hz bandwidth expansion: $w_{\text{lag}}(i) = e^{-(2\pi f_0 i/f_s)^2/2}, i = 1, \ldots, M$, where f_s is the sampling frequency.

Bandwidth expansion can also be obtained by scaling the radii of the poles of the all-pole model inwards with a factor $\gamma < 1$ [13]. The scaling is done by multiplying the LP coefficients $a_i' = \gamma^i a_i$, and the desired bandwidth expansion f_0 determines the constant via the approximate expression $\gamma \approx e^{-\pi f_0/f_s}$ [6], [14].

The method of bandwidth expansion via radii scaling of the poles is used for example in the code excited linear prediction (CELP) codec in MPEG-4 [15] that uses the factor $\gamma = 0.9902$ for 8-kHz sampling rate and $\gamma = 0.9883$ for 16-kHz sampling rate, corresponding to 25- and 60-Hz bandwidth expansion, respectively. Another example is the ITU-T G.728 [16], which uses $\gamma = 253/256$ at 8-kHz sampling frequency to obtain 30 Hz of bandwidth expansion.

The popularity of the described methods of bandwidth expansion is mainly due to their simplicity. A small modification to LP, either to the autocorrelation sequence or directly to the coefficients of the AR model, makes the spectral envelope smoother, which leads to improved performance of speech coders, speaker recognition and speech modification systems.

Smoothing *regularization* by penalizing an approximate first- or second-derivative of the model (i.e., envelope in our case) is a common approach in optimization to control a tradeoff between fit of data and model smoothness [17]. In [18], the authors proposed a regularization technique to minimize a composite error criterion for discrete cepstrum estimation. The work was continued in [19] and [20], where the regularization approach was applied towards the reduction of the peaky behavior of LP-based spectral estimation. The main disadvantage of the solution of [19] and [20] is the use of an iterative optimization method, leading to significantly higher complexity than that of conventional LP methods.

Murthi and Kleijn [21] introduced an approximation of the smoothness criterion to eliminate the need for an iterative optimization procedure in the calculation of the AR coefficients. A proper choice of the regularization parameter λ was left open. The choice of the regularization parameter was investigated in [22] for the narrowband case and for model order 16 only.

In this paper, we fill in the gaps of previous work and complete our investigation on regularized LP. We provide a heuristic approach to investigate the stability of regularized LP. We further present extensive results on selecting the regularization parameter λ for a range of model orders and for both the narrow- and wide-band case, and also present comparisons to conventional LP methods. These experiments clearly show that spectral

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estimates based on regularized LP are better than those based on conventional LP methods at an insignificant additional computational complexity.

The remainder of this paper is organized as follows. Section II describes the theory of regularized LP. In Section III, we describe our simulations and determine the optimal regularization parameter and introduce models for adaptive regularization. In Section IV, we validate our results from the simulations and investigate the outliers of the spectral estimation errors. Section V contains our conclusions, and in the Appendix we investigate the stability of regularized LP.

II. Theory of Regularized LP

In LP, the AR model coefficients are estimated by minimizing the variance of the prediction error signal. We call this cost criterion $\mathcal{D}$. Regularized LP includes a penalty measure $\mathcal{R}$ that increases as the spectral envelope gets peakier. The composite cost function used is $\mathcal{D} + \lambda\mathcal{R}$, where the regularization constant λ controls the tradeoff between the fit to data (i.e., the minimum prediction error variance $\mathcal{D}$ as in LP) and the smoothness of the envelope. As λ increases, more emphasis is put on the smoothness criterion and the envelope gets smoother. As $\lambda \to 0$, the penalty measure is ignored and the method reduces to conventional LP.

A. Linear Prediction

For an AR model $1/A(z)$ of order M with $A(z) = 1 + a_1 z^{-1} + \cdots + a_M z^{-M}$, let the spectral envelope be defined by

$$S(\omega, \mathbf{a}) = \frac{1}{|A(e^{j\omega})|^2} \tag{1}$$

in which $\mathbf{a} = [a_1, a_2, \ldots, a_M]^T$ denotes the AR model coefficients. In LP, these coefficients are determined through the minimization of the variance of the prediction error signal $e(n) = \sum_{k=0}^{M} a_k s(n-k)$, where $s(n)$ are the speech samples, and $a_0 = 1$. The cost criterion of LP can thus be written as

$$\mathcal{D} = E[e^2(n)] = \mathbf{a}^T\mathbf{R}\mathbf{a} + 2\mathbf{a}^T\mathbf{r} + r_0 \tag{2}$$

where

$$\mathbf{R} = \begin{bmatrix} r_0 & r_1 & \cdots & r_{M-1} \\ r_1 & r_0 & & r_{M-2} \\ \vdots & & \ddots & \\ r_{M-1} & r_{M-2} & \cdots & r_0 \end{bmatrix} \tag{3}$$

$$\mathbf{r} = [r_1,\ r_2,\ \ldots, r_M]^T \tag{4}$$

and $r_k = E[s(n)s(n-k)]$ are the autocorrelation values of the speech signal. Minimizing (2) with respect to the AR model coefficients $\mathbf{a}$ gives the solution

$$\mathbf{a}_{\text{opt}} = -\mathbf{R}^{-1}\mathbf{r} \tag{5}$$

which usually is solved through the computationally efficient Levinson–Durbin recursion [23], [24].

B. Regularization

In regularized LP, rapid changes in the contour of the spectral envelope are penalized through a penalty measure. In [18] and [19], the authors suggest the penalty measure

$$\mathcal{R}(S(\omega, \mathbf{a})) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\left[\frac{d}{d\omega}\log S(\omega, \mathbf{a})\right]^2 d\omega \tag{6}$$

which increases with increasing peakiness of the spectral envelope. The criterion in (6) has the property that minimizing $\mathcal{D} + \lambda\mathcal{R},\ \lambda \geq 0$, where $\mathcal{D}$ is given by (2), leads to a stable all-pole filter $1/A(z)$ (proof in the Appendix). The criterion (6) is difficult to incorporate into noniterative estimation methods to find $\mathbf{a}_{\text{opt}}$, and in [21] an approximation to the penalty measure is proposed as follows:

$$\frac{1}{2\pi}\int_{-\pi}^{\pi}\left|\frac{d}{d\omega}\log(A(e^{j\omega}))\right|^2 d\omega = \frac{1}{2\pi}\int_{-\pi}^{\pi}\left|\frac{A'(e^{j\omega})}{A(e^{j\omega})}\right|^2 d\omega \tag{7}$$

where $A'(e^{j\omega})$ denotes the derivative with respect to ω. The denominator in (7) still prevents a noniterative solution to the problem. Therefore, the penalty measure is modified further to

$$\widehat{\mathcal{R}}(S(\omega, \mathbf{a})) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\left|\frac{A'(e^{j\omega})}{W(\omega)}\right|^2 d\omega \tag{8}$$

where $1/|W(\omega)|^2$ serves as a coarse approximation of the spectral envelope $1/|A(e^{j\omega})|^2$. Note that the stability of $1/A(z)$ is not proven when (8) is used, but we have not observed any unstable filters in practice.

Since $A'(e^{j\omega}) = -\sum_{k=0}^{M} jka_k e^{jk\omega}$, the expression in (8) can be written as

$$\sum_{k=0}^{M}\sum_{l=0}^{M} ka_k la_l \frac{1}{2\pi}\int_{-\pi}^{\pi}\frac{e^{-j\omega(k-l)}}{|W(\omega)|^2}d\omega. \tag{9}$$

The integral in (9) is the inverse discrete time Fourier transform of $|1/W(\omega)|^2$ and we can write

$$\widehat{\mathcal{R}}(S(\omega, \mathbf{a})) = \sum_{k=0}^{M}\sum_{l=0}^{M} ka_k la_l f(l-k) \tag{10}$$

where

$$f(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\frac{e^{j\omega x}}{|W(\omega)|^2}d\omega. \tag{11}$$

To give $1/|W(\omega)|^2$ a coarse spectral envelope structure, let $f(m) = r(m)v(m)$ be the autocorrelation sequence of the speech signal windowed with a window $v(m)$. This means that $1/|W(\omega)|^2$ is a convolution of the speech power spectrum described by the length M autocorrelation sequence $r(m)$ with the window $V(\omega)$.

Equation (10) can be expressed as a quadratic cost function

$$\widehat{\mathcal{R}}(S(\omega, \mathbf{a})) = \mathbf{a}^T\mathbf{D}\mathbf{F}\mathbf{D}\mathbf{a} \tag{12}$$

where $\mathbf{D}$ is a diagonal matrix in which each diagonal element consists of the row number, and $\mathbf{F}$ is a Toeplitz autocovariance matrix for the windowed autocorrelation sequence $f(m)$. We

create the composite cost function $\mathcal{D} + \lambda\widehat{\mathcal{R}}$ using (2) and (12) to get

$$\mathbf{a}^T(\mathbf{R} + \lambda\mathbf{DFD})\mathbf{a} + 2\mathbf{a}^T\mathbf{r} + r_0 \tag{13}$$

which gives the compact solution

$$\mathbf{a_{opt}} = -(\mathbf{R}+\lambda\mathbf{DFD})^{-1}\mathbf{r}. \tag{14}$$

The matrix $(\mathbf{R} + \lambda\mathbf{DFD})$ is Hermitian and positive definite, so efficient algorithms for solving (14) exist, e.g., Cholesky decomposition [25]. Although these methods are not as efficient as the Levinson–Durbin recursion, the additional computational complexity over conventional LP methods is negligible in most applications.[1]

III. Selection of λ

The open issue of regularized LP is to choose the regularization parameter λ, and to make performance comparisons to conventional LP methods. In this paper, we investigate two approaches for choosing λ.

1) *Constant* λ. In this case, λ is kept constant throughout every frame and for all speakers. This approach is simple to implement and use, and the added computational complexity over conventional LP methods is often negligible. We need to find the best constant value of λ.
2) *Adaptive* λ. In this case, we allow λ to vary on a frame-by-frame basis, speaker-by-speaker, or by gender. To use an adaptive approach for selecting λ, we would like a model that selects the best λ given the present conditions, e.g., speaker gender or pitch frequency of the current frame.

The window $v(m)$ is chosen as a boxcar window of the same length as model order, so that the special autocorrelation matrix $\mathbf{F} = \mathbf{R}$, which is a coarse spectral envelope of the speech frame.

We choose to compare regularized LP to bandwidth-expanded LP, where the poles are scaled inwards with a factor γ. The comparison becomes straightforward, and the bandwidth expansion parameter γ can be tuned to minimize a quality criterion in the same way as λ to get a fair comparison. In our experience, lag windowing of the autocorrelation sequence has a similar effect on the envelope as bandwidth expansion, and so we omit lag windowing from the simulation setup. The iterative method suggested in [19] and [20] is not used for comparison, because the authors do not provide a description of a method to implement regularization for the case of linear prediction. The different envelope estimation methods investigated are summarized in Table I.

In determining methods for selecting λ, we need to gauge the envelope modeling quality. To proceed, we outline a method for generating a reference envelope from a frame of speech, with the reference envelope serving as an ideal model of the unknown vocal tract transfer function. The validity of the modeling efficacy of the reference envelope is demonstrated through a controlled experiment utilizing synthetic voiced speech. Then we examine the performance of regularized LP and bandwidth-expanded LP in modeling the various reference envelopes determined from a real speech training set to pinpoint the choice of λ.

[1]For a sampling frequency of 16 kHz, using model order 16 and a 20-ms speech frame, our nonoptimized Matlab implementation of regularized LP needs approximately 0.80 ms to execute on a Intel Core 2 Duo 2 GHz, whereas Matlab's implementation of linear prediction finishes in 0.38 ms.

TABLE I
Approaches of Selecting λ and γ

Constant	$\mathbf{LP}$	Linear Prediction is used as a reference method.
	$\mathbf{R_C}$	Regularized LP with the constant λ that yields the lowest distortion.
	$\mathbf{BE_C}$	Bandwidth-expanded LP with the constant γ that yields the lowest distortion.
Adaptive	$\mathbf{R_A}$	Regularized LP with adaptive λ, chosen as a function of the pitch frequency.
	$\mathbf{R_{opt}}$	Regularized LP with λ changing on a frame-by-frame basis, picked as the λ that gives the smallest distortion possible for each particular frame.
	$\mathbf{BE_{opt}}$	Bandwidth-expanded LP with γ changing on a frame-by-frame basis, picked as the γ that gives the smallest distortion possible for each particular frame.

For training and validation data, we use voiced frames from the TIMIT database [26], sampled at 16 kHz. 9354 female frames and 7498 male frames were used for training, and another 3926 female frames and 4430 male frames from other speakers for validation. The frames were windowed with a 20-ms Hamming window prior to analysis. The frames were sampled at both 8 and 16 kHz, and the performance of models at order 10, 16, 22, and 28 were investigated.

A. Objective Quality Criterion

The performance of a spectral envelope estimation method can be measured in many ways. An often used criterion for measuring quality is the spectral distortion at the harmonics between the power spectrum of the speech frame and the modeled spectral envelope (e.g., [7], [27]–[29]). In our case, this measure is less suitable because it does not indicate whether the AR model of the envelope is contaminated by the spectral fine structure (the harmonics). Instead, we would like to evaluate the envelope modeling performance as the spectral distortion between the true vocal tract transfer function and the spectral envelope of the AR models over the entire frequency support. Since we cannot obtain the true vocal tract transfer function, we use the cubic spline interpolation between the harmonic peaks of the logarithmic periodogram as a coarse approximation of the true vocal tract transfer function and, hence, obtain a reference envelope. We calculate the log spectral distortion between our reference envelope $S_{\text{int}}(\omega)$, and the estimated AR model $S(\omega, \mathbf{a})$ as

$$\text{SD} = \sqrt{\frac{1}{2\pi}\int_{-\pi}^{\pi}\left[10\log_{10} S_{\text{int}}(\omega) - 10\log_{10} S(\omega, \mathbf{a})\right]^2 d\omega}. \tag{15}$$

Although it would be possible to obtain an LP approximation from the reference envelope, that would involve complex calculations (e.g., periodogram estimation, finding peaks, cubic spline interpolation) and is therefore not practical as an efficient LP envelope estimation method.

Fig. 1 shows an example of the reference envelope for a voiced frame of a female speaker. The plot also shows the envelope of LP and regularized LP (shifted down for better visibility). In this example, the parameter of regularization

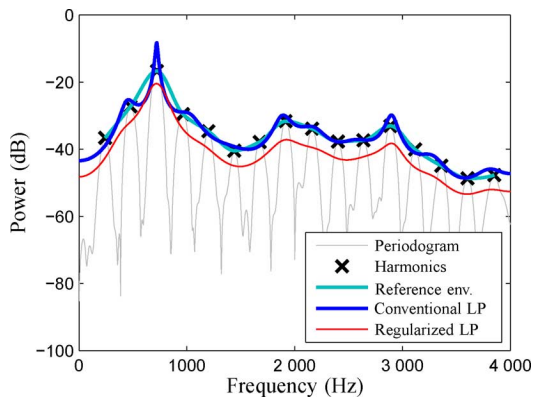

Fig. 1. Periodogram of a voiced frame of female speech (solid gray), using a 20-ms Hamming window. The harmonics and the reference envelope (cubic spline interpolation $S_{\rm int}$) are marked in the figure. The conventional LP envelope estimate of order 22 is shown as a dark bold curve, and the envelope of regularized LP of the same model order is shown as a solid curve that is shifted downwards for better visibility. The optimal λ in this case is 3.28×10^{-3}, which gives a smooth envelope that very closely resembles the reference envelope.

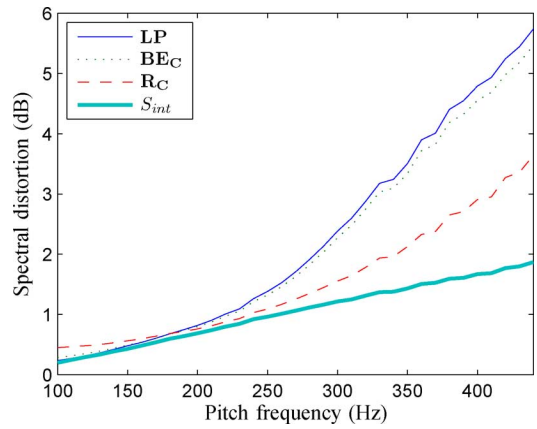

Fig. 2. Spectral distortion for the different envelope estimation methods using a known AR model as reference. The cubic spline interpolation between the harmonics of the periodogram (bold curve) has a lower average spectral distortion than the LP-based envelope estimation methods for all pitch frequencies, and thus is a better approximation of the true spectral envelope. However, $S_{\rm int}$ is impractical, and the regularization approach is the best choice of the LP-based envelope estimation methods.

was chosen to minimize the spectral distortion in (15), and the envelope of the regularization approach closely resembles the reference envelope.

1) Appropriateness of the Reference Envelope: To verify that the reference envelope used in (15) is a good approximation of the true envelope, a set of known smooth AR models was generated by using bandwidth-expanded LP on the set of frames from the validation data set. In particular, for each frame of real speech from the validation data, bandwidth-expanded LP was used for estimating an AR model. We then created synthetic speech by using a pulse train of a particular pitch frequency to excite the AR model, which was considered as the true spectral envelope. With the time-domain synthetic speech signal, we calculated spectral envelope estimates utilizing the different estimation methods ($\mathbf{LP}$, $\mathbf{BE_C}$, $\mathbf{R_C}$), and the reference envelope method. Then, we calculated the spectral distortion between the fixed AR model, and various estimated spectral envelopes. This controlled experiment was repeated for various pitch frequencies between 100 and 450 Hz with 10-Hz spacing and for each frame in the validation database. The results were consistent over all model orders and sampling frequencies; the spectral distortions of the reference envelope was always smaller than the distortions of the LP-based estimates for all pitch frequencies. Thus, our reference envelope is a better approximation to the true envelope than the LP-based envelope estimation methods. Fig. 2 shows the spectral distortion averaged over all envelopes and all modes (female, male, sampling frequency, model order). For the pitch estimation and the peak-picker for the periodogram, the Matlab speech processing toolbox Voicebox [30] was used.

B. Finding the Optimal λ

For each frame of the training database, we calculated a reference envelope using the cubic spline interpolation approach. We then estimated the spectral envelope using the various LP-based methods based on the real speech frame, sweeping over different values of λ and γ. The spectral distortion measure defined in

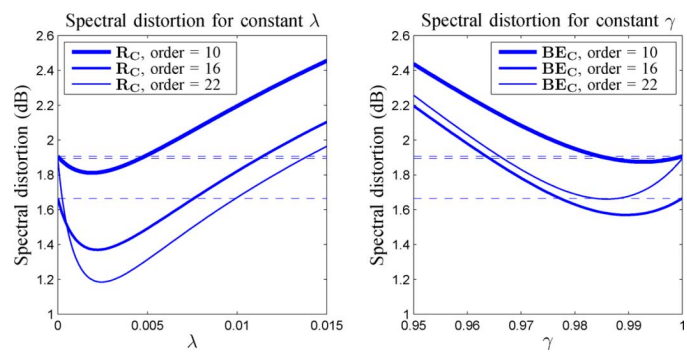

Fig. 3. Average spectral distortion over the 9354 female frames of the training database for 8-kHz sampling frequency. Model orders 10, 16, and 22 are used, and the dashed lines show the distortion of conventional LP ($\lambda \rightarrow 0, \gamma \rightarrow 1$) for the different model orders.

(15) was calculated for a range of different parameter values to find the optimal λ and γ for the methods given by Table I. The optimal value of λ was refined using the Newton method [31].

1) Constant λ: As a first example, the spectral distortion from the simulation with the models with constant λ and γ ($\mathbf{R_C}$ and $\mathbf{BE_C}$) for female speakers and 8-kHz sampling frequency is shown in Fig. 3. From the figure, it is clear that there is a minimum in the spectral distortion, which is the best constant values of λ and γ that we can choose for this particular case. The other cases (female 16 kHz, male 8 and 16 kHz) have plots that are consistent in shape and the results are summarized in Table II. It is seen that regularized LP is better than bandwidth-expanded LP for all model orders, and that the spectral distortion performance of regularized LP improves with increasing model order. Also note that the best constant λ tends to increase with increasing model order when the training is performed on both female and male frames simultaneously, as shown by the bold lines in Fig. 4 and Table III. For γ, the opposite holds true, which agrees with intuition. The spectral distortions of conventional LP is shown as horizontal dashed lines in Fig. 3, and from this

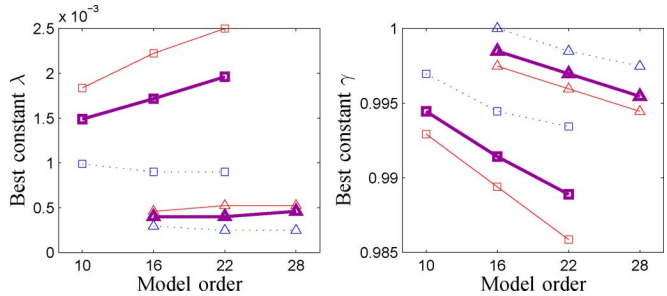

Fig. 4. Best constant λ and γ as a function of model order for female (solid lines) and male (dotted lines) speakers and sampling frequencies 8 kHz (squares) and 16 kHz (triangles). The bold lines show the result from training with both female and male speakers (numbers given in Table III).

TABLE II
SPECTRAL DISTORTION (IN dB) OF LP AND THE DECREASE IN SD FOR THE DIFFERENT SPECTRAL ESTIMATION METHODS

		$f_s = 8$ kHz			$f_s = 16$ kHz		
	Order	10	16	22	16	22	28
Female	SD of $\mathbf{LP}$	1.91	1.67	1.89	2.42	1.96	1.82
	Decr. $\mathbf{BE}_{\mathbf{opt}}$	0.050	0.124	0.274	0.017	0.041	0.074
	Decr. $\mathbf{BE}_{\mathbf{C}}$	0.032	0.096	0.233	0.007	0.026	0.056
	Decr. $\mathbf{R}_{\mathbf{opt}}$	0.129	0.350	0.784	0.060	0.104	0.181
	Decr. $\mathbf{R}_{\mathbf{C}}$	0.095	0.296	0.709	0.039	0.076	0.145
	Decr. $\mathbf{R}_{\mathbf{A}}$	0.102	0.318	0.747	0.043	0.084	0.157
Male	SD of $\mathbf{LP}$	2.44	1.99	1.85	3.22	2.63	2.44
	Decr. $\mathbf{BE}_{\mathbf{opt}}$	0.031	0.057	0.079	0.016	0.035	0.052
	Decr. $\mathbf{BE}_{\mathbf{C}}$	0.007	0.023	0.040	0.000	0.004	0.012
	Decr. $\mathbf{R}_{\mathbf{opt}}$	0.076	0.101	0.150	0.045	0.059	0.072
	Decr. $\mathbf{R}_{\mathbf{C}}$	0.035	0.051	0.093	0.016	0.025	0.033
	Decr. $\mathbf{R}_{\mathbf{A}}$	0.040	0.058	0.101	0.018	0.028	0.037

The decrease in spectral distortion shows that regularized LP consistently outperforms bandwidth-expanded LP. The results are from the training database.

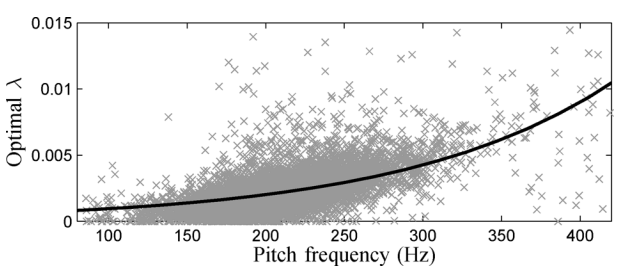

Fig. 5. Optimal λ as a function of the pitch frequency for the female frames of the training database (x) at 8 kHz and model order 16. The curve shows the best fit to the data of the exponential model in (16).

we gather that the optimal model order for LP lies closer to 16 than 10 or 22, for female narrowband speech.

2) Adaptive λ: For each frame, the optimal λ is plotted against the pitch frequency in Fig. 5 for the case of female speakers at model order 16 with sampling frequency 8 kHz. The plot shows a clear correlation between the optimal value of λ and the pitch frequency of the frame, which can be used in an adaptive method. The adaptive model for λ that we propose is

$$\lambda(f_p) = ae^{bf_p} \tag{16}$$

where f_p denotes the pitch frequency, and $a > 0$. The adaptive model is also shown in Fig. 5 as a solid curve. The modeling was repeated for every case and the best model parameters in a least-square sense are presented in Table III. The simulation results showed that the best model parameters for female and male speech were very similar, so the simulation results were grouped together to get only one set of model parameters. The conclusion is that the optimal value of λ is dependent on pitch, not gender. That conclusion is beneficial when using regularized LP in a speech application, where the gender of the current speaker seldom is known. The performance of the adaptive method ($\mathbf{R}_{\mathbf{A}}$) is consistently better than the method of constant λ ($\mathbf{R}_{\mathbf{C}}$) which can be seen in Table II. In the methods $\mathbf{R}_{\mathrm{opt}}$ and $\mathbf{BE}_{\mathrm{opt}}$, the λ and γ that minimizes (15) are chosen for each frame. They are included to determine the optimal performance of the methods of regularized LP and bandwidth-expanded LP, respectively, and cannot be used in practical applications. The relatively poor performance of $\mathbf{BE}_{\mathrm{opt}}$ show that even with an optimized model function for γ, the method of bandwidth-expanded LP cannot compete with the regularization methods.

TABLE III
BEST CONSTANT λ AND PARAMETERS OF THE EXPONENTIAL MODEL

	$f_s = 8$ kHz			$f_s = 16$ kHz		
Order	10	16	22	16	22	28
$\lambda_{\mathbf{R}_{\mathbf{C}}}$ ($\times 10^{-3}$)	1.49	1.72	1.96	0.40	0.40	0.46
a ($\times 10^{-4}$)	5.73	4.46	4.55	1.49	1.12	1.07
b ($\times 10^{-3}$)	5.73	7.50	8.00	5.61	7.24	7.96

3) λ Parameter Sensitivity: The sensitivity to the suggested constant values and models of λ were investigated in terms of window length. Hamming windows of size 20 and 40 ms were used and compared, and we concluded that the difference in optimal values of λ and in the model parameters of $\mathbf{R}_{\mathbf{A}}$ was insignificant.

IV. VALIDATION

In this section, we validate our results from the previous section with validation data taken from different speakers than those used in the training data. This section also covers the topic of spectral estimation error outliers, where we investigate the large spectral estimation errors of regularized LP and the conventional LP methods.

A. Spectral Distortion

We used methods $\mathbf{R}_{\mathbf{C}}$, $\mathbf{R}_{\mathbf{A}}$, and $\mathbf{BE}_{\mathbf{C}}$ with the best values for λ and γ and model parameters obtained through the training session without the distinction between female and male speakers (summarized in Table III). The validation results are consistent with the results from the training section and are presented in Fig. 6. Regularized LP has a clear benefit over bandwidth-expanded LP. The advantage of using regularized LP increases with increasing model order, which is expected since conventional LP methods suffer from harmonic contamination of the spectral envelope at higher model orders.

The pitch frequency dependency is clear from Fig. 7, which shows the decrease in spectral distortion for the three methods divided over pitch frequency bins for the case of narrow-band speech. The benefit of using the adaptive model is clear for the higher pitch frequencies in this figure. The 95% confidence intervals shown in Fig. 7 are based on the Student's t-distribution (e.g., [32]) and are to be considered approximate for the very

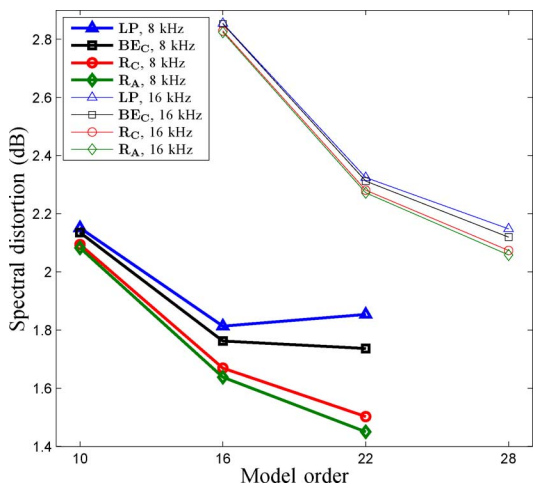

Fig. 6. Average spectral distortion on the validation data set. The figure clearly shows the benefits of regularized LP compared to bandwidth-expanded LP.

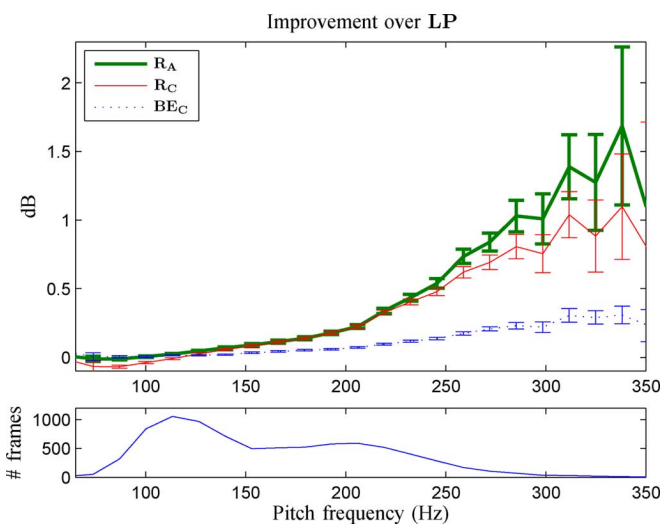

Fig. 7. Decrease in spectral distortion over LP at different pitch frequencies, for the investigated methods. The error bars in the upper plot show the 95% confidence intervals. The lower plot shows the distribution of the frames over pitch frequency. The results are from the validation database, with sampling frequency 8 kHz and model order 16.

highest pitch frequencies due to the small number of speech frames at those frequencies.

B. Spectral Estimation Error Outliers

The spectral estimation error outliers are studied to see if the regularized LP approach gives a lower frequency of occurrence of large spectral errors when compared to LP with bandwidth expansion. This would mean that the regularized LP method is less prone to giving audible artifacts in coding and other speech processing systems.

The approach used was similar to that of Section III-A1, in that we performed a controlled experiment on synthetic speech of various pitch frequencies and fixed, known AR models. The validation database was used, and the l_∞ norm of the log spectral distortion between the estimated envelopes and the true envelopes was measured. The l_∞ norm is the maximum absolute

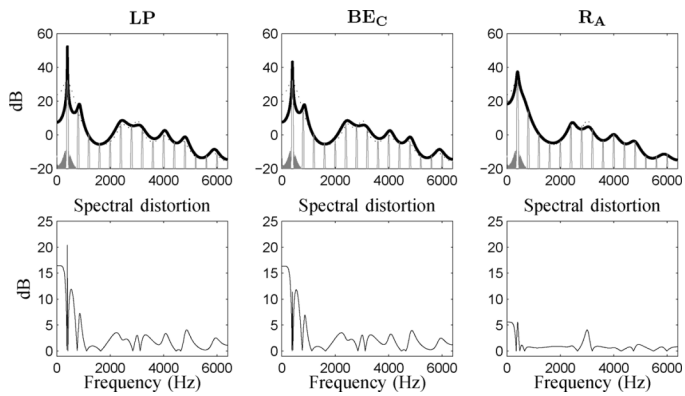

Fig. 8. Example of one frame, with a pitch frequency of 400 Hz and $f_s = 12.8$ kHz. In the upper plots, the periodogram is shown with the solid gray curve, the true AR model with a dotted curve, and the different LP estimation methods are shown in thick solid curves. In the lower plots, the absolute spectral distortion of the same envelopes are shown for the different methods. Note that the regularization method has a low distortion at the low frequency bands, which is a desirable property as auditory perception is particularly acute in that frequency region.

difference between the true envelope $S(\omega, \mathbf{a})$ and the LP-based envelope estimate $S(\omega, \hat{\mathbf{a}})$

$$\|\mathrm{SD}\|_\infty = \max_\omega |10 \log_{10} S(\omega, \mathbf{a}) - 10 \log_{10} S(\omega, \hat{\mathbf{a}})|$$

and is used to catch the largest outliers in the envelope estimates. The parameters in $\hat{\mathbf{a}}$ are estimated for all four LP-based envelope estimation methods ($\mathbf{LP}$, $\mathbf{BE_C}$, $\mathbf{R_C}$, $\mathbf{R_A}$).

The tests were conducted for both narrow-band and wide-band speech. In the case of narrow-band speech, the sampling frequency was 8 kHz and the model order was 10, and for wide-band speech, a sampling frequency of 12.8 kHz and order 16 was used to mimic AMR [11] and AMR-WB [12]. The parameter values were taken from the training summarized in Table III, and wide-band were considered to be 16 kHz with model order 16.

Before providing the statistical results we provide an illustrative example. Fig. 8 shows an example frame, at a pitch frequency of approximately 400 Hz, sampling frequency 12.8 kHz, and model order 16. We can see that conventional LP gives an unreasonably sharp peak at the first harmonic. Bandwidth expansion reduces the peak but still suffers from harmonic contamination. Regularized LP has a very smooth envelope with a small spectral distortion.

For each one of the 8356 envelopes of the validation database, we obtained l_∞ norm distortion values with respect to pitch frequency for the four different envelope estimation methods. Percentiles were taken of the l_∞ norm distortion values, and the result is shown in Table IV.

The regularized LP methods have lower distortion than the others at the higher percentiles, which shows that regularized LP is less prone to large spectral estimation errors. Regularized LP gives fewer large spectral estimation errors than bandwidth-expanded LP, which is clearly beneficial in applications where error outliers must be avoided, such as in speech modification and speech coders that do not use analysis-by-synthesis, e.g., vocoders. The method with adaptive λ ($\mathbf{R_A}$) performs better

TABLE IV
PERCENTILES (P) OF THE l_∞ NORM OF THE SPECTRAL ESTIMATION ERRORS (IN dB)

	Narrowband				Wideband			
P	$\mathbf{LP}$	$\mathbf{BE_C}$	$\mathbf{R_C}$	$\mathbf{R_A}$	$\mathbf{LP}$	$\mathbf{BE_C}$	$\mathbf{R_C}$	$\mathbf{R_A}$
5	0.81	0.83	0.88	0.79	0.81	0.79	0.72	0.70
25	2.18	2.10	2.06	1.95	1.83	1.79	1.70	1.67
50	4.46	4.06	3.91	3.58	3.75	3.60	3.40	3.20
75	7.45	6.59	6.20	5.73	6.52	6.18	5.72	4.98
90	11.52	9.06	8.18	7.88	10.98	9.72	8.26	6.83
95	14.94	10.82	9.42	9.14	14.22	11.50	9.51	7.98
99	19.19	14.73	12.10	11.29	18.33	15.70	12.75	10.38

The regularization methods have smaller spectral estimation error outliers, meaning that they are less prone to large spectral errors.

than constant λ ($\mathbf{R_C}$). The very large spectral estimation errors mainly occur at the highest pitch frequencies. In this experiment, bandwidth-expanded LP might benefit from the fact that we are using the same method to obtain the true envelopes. Since the results are still favorable for regularized LP, the conclusion is simple—regularized LP does give fewer large spectral estimation errors.

V. CONCLUSION

Regularized linear prediction yields a smoother and physically more plausible spectral envelope than linear prediction with bandwidth expansion. Whereas conventional linear prediction methods simply modify an existing LP estimate of the spectral envelope that may be contaminated by spectral fine structure, regularized LP optimizes the spectral fit with a penalty on nonsmooth behavior. This provides the regularization procedure with an advantage over conventional LP methods at a fundamental level. As a result, regularized LP can provide an accurate estimate of the spectral envelope with a high model order.

Our simulations show that the regularized linear prediction models an approximation to the vocal tract transfer function more accurately than bandwidth-expanded linear prediction, even for a constant penalty factor λ. We propose a model for an adaptive method using a pitch estimate of the current frame, which increases the performance further. Our simulations also show that regularized LP has fewer large spectral estimation errors than conventional LP methods.

Regularized LP provide benefits over conventional LP in speech applications such as 1) speech modification, 2) speech coding, and 3) other applications where it is of importance to find a good estimate of the true spectral envelope. In most applications, regularized linear prediction has a negligible additional computational cost over conventional LP methods. Thus, regularized linear prediction forms an attractive method for estimation of the AR parameters in speech processing systems.

APPENDIX

In conventional LP, the prediction error variance is minimized. Denote the minimization criterion $\mathcal{D}(\mathbf{a})$ so that

$$\dot{\mathbf{a}} = \arg\min_{\mathbf{a}} \mathcal{D}(\mathbf{a}) \tag{17}$$

is the coefficient vector of LP. In regularized LP, we instead minimize $\mathcal{D}(\mathbf{a}) + \lambda\mathcal{R}(\mathbf{a})$. Let

$$\tilde{\mathbf{a}} = \arg\min_{\mathbf{a}} \mathcal{D}(\mathbf{a}) + \lambda\mathcal{R}(\mathbf{a}) \tag{18}$$

be the corresponding solution for regularized LP, for some $\lambda > 0$. From (17) and (18), it is obvious that

$$\mathcal{D}(\dot{\mathbf{a}}) < \mathcal{D}(\tilde{\mathbf{a}}) \tag{19}$$

$$\mathcal{D}(\tilde{\mathbf{a}}) + \lambda\mathcal{R}(\tilde{\mathbf{a}}) \leq \mathcal{D}(\dot{\mathbf{a}}) + \lambda\mathcal{R}(\dot{\mathbf{a}}) \tag{20}$$

where in (19) we assumed that $\mathcal{R}(\mathbf{a})$ is strictly positive, i.e., that the envelope is not flat. Combining (19) and (20) gives

$$\mathcal{R}(\tilde{\mathbf{a}}) \leq \frac{\mathcal{D}(\dot{\mathbf{a}}) - \mathcal{D}(\tilde{\mathbf{a}})}{\lambda} + \mathcal{R}(\dot{\mathbf{a}}) < \mathcal{R}(\dot{\mathbf{a}}) \tag{21}$$

which shows that $\mathcal{R}(\mathbf{a})$ decreases as the regularization criterion is used for any $\lambda > 0$.

We now show that the integral (6) increases when the least stable pole is scaled outwards. We can rewrite (6) as

$$\mathcal{R}(\mathbf{a}) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\left[\frac{-1}{|A(e^{j\omega})|^2}\frac{d}{d\omega}|A(e^{j\omega})|^2\right]^2 d\omega. \tag{22}$$

Let $A(e^{j\omega}) = \prod_{i=1}^{M}(1 - r_i e^{-j(\omega-\alpha_i)})$ define the AR model ($r_i \in [0,1)\ \forall\ i$). It then follows that

$$|A(e^{j\omega})|^2 = \prod_{i=1}^{M}\left(1 - 2r_i\cos(\omega-\alpha_i) + r_i^2\right)$$

$$\frac{d}{d\omega}|A(e^{j\omega})|^2 = \sum_{i=1}^{M}\frac{2r_i\sin(\omega-\alpha_i)}{1 - 2r_i\cos(\omega-\alpha_i) + r_i^2}|A(e^{j\omega})|^2$$

and, thus, the integral (22) can be written as

$$\frac{1}{2\pi}\int_{-\pi}^{\pi}\left(\sum_{i=1}^{M}\frac{2r_i\sin(\omega-\alpha_i)}{1 - 2r_i\cos(\omega-\alpha_i) + r_i^2}\right)^2 d\omega. \tag{23}$$

Without loss of generality, we can assume that the poles are sorted, i.e., $r_1 > r_2 > \cdots > r_M$, so that $r_1 e^{j\alpha_1}$ corresponds to the least stable pole of $1/|A(e^{j\omega})|$. The derivative of (23) with respect to r_1 is

$$\frac{8}{2\pi}\int_{-\pi}^{\pi} g(\omega)h(\omega)d\omega \tag{24}$$

where

$$g(\omega) = \frac{1 - r_1^2}{(1 - 2r_1\cos(\omega-\alpha_1) + r_1^2)^2} \tag{25}$$

$$h(\omega) = \sum_{i=1}^{M}\frac{r_i\sin(\omega-\alpha_i)\sin(\omega-\alpha_1)}{1 - 2r_i\cos(\omega-\alpha_i) + r_i^2} \tag{26}$$

and where we moved the differentiation under the integral sign. This is allowed according to Leibniz's integral rule [33] since both the integrand of (23) and the integrand of (24) are continuous on $\omega \in [-\pi,\pi],\ r_i \in [0,1),\ i = 1,\ldots,M$. The function $g(\omega) \geq 0$ takes its maximum value $(1+r_1)/(1-r_1)^3$ at $\omega = \alpha_1$, and is symmetric around that point. We divide the integral (24)

into three parts: $\int_{-\pi}^{\alpha_1-\epsilon} + \int_{\alpha_1-\epsilon}^{\alpha_1+\epsilon} + \int_{\alpha_1+\epsilon}^{\pi} = I_1 + I_2 + I_3$, for some $\epsilon > 0$. We can write

$$\begin{aligned} h(\omega) &= \frac{r_1 \sin^2(\omega-\alpha_1)}{1-2r_1\cos(\omega-\alpha_1)+r_1^2} \\ &\quad + \sum_{i=2}^{M} \frac{r_i \sin(\omega-\alpha_i)\sin(\omega-\alpha_1)}{1-2r_i\cos(\omega-\alpha_i)+r_i^2} \\ &\geq 0 - \sum_{i=2}^{M} \frac{r_i}{(1-r_i)^2} \geq \frac{-(M-1)r_{\max}}{(1-r_{\max})^2} \equiv -C \end{aligned}$$

where C is a bounded positive constant, since $r_{\max} = \max\{r_i\}_{i=2}^{M} < 1$. We can thus write

$$\begin{aligned} I_1 + I_3 &\geq -\frac{4C}{\pi}\left(\int_{-\pi}^{\alpha_1-\epsilon} g(\omega)d\omega + \int_{\alpha_1+\epsilon}^{\pi} g(\omega)d\omega\right) \\ &\geq -\frac{4C}{\pi}(2\pi-2\epsilon)g(\alpha_1+\epsilon) \\ &= -\frac{4C\left(2-\frac{2\epsilon}{\pi}\right)(1-r_1^2)}{(1-2r_1\cos(\epsilon)+r_1^2)^2} \end{aligned}$$

where we used that $g(\omega)$ is symmetric around α_1 and takes its largest value in the interval $[-\pi, \alpha_1-\epsilon) \cup (\alpha_1+\epsilon, \pi]$ at $\omega = \alpha_1 \pm \epsilon$. $I_1 + I_3$ is lower bounded by a number that approaches 0 as $r_1 \to 1^-$, which proves that for a sufficiently large r_1, we have that $I_1 + I_3 \geq 0$. In the middle interval, $\omega \in [\alpha_1-\epsilon, \alpha_1+\epsilon]$, we have

$$\begin{aligned} h(\omega) &= \frac{r_1(1-\cos(\omega-\alpha_1))(1+\cos(\omega-\alpha_1))}{2r_1(1-\cos(\omega-\alpha_1))+(1-r_1)^2} \\ &\quad + \sum_{i=2}^{M} \frac{r_i \sin(\omega-\alpha_1)\sin(\omega-\alpha_i)}{1-2r_i\cos(\omega-\alpha_i)+r_i^2} \\ &\geq \frac{1+\cos(\epsilon)}{2+\frac{(1-r_1)^2}{r_1(1-\cos(\omega-\alpha_1))}} - \sin(\epsilon)\sum_{i=2}^{M}\frac{r_i}{(1-r_i)^2} \\ &\geq \frac{2-\epsilon}{2+\frac{(1-r_1)^2}{r_1(1-\cos(\omega-\alpha_1))}} - \epsilon C \\ &\to 1 - \frac{\epsilon}{2} - \epsilon C, \quad \text{as} \quad r_1 \to 1^- \end{aligned}$$

which is strictly positive if $\epsilon < 2/(1+2C)$. We can pick an $\epsilon \in (0, 2/(1+2C))$ such that $I_2 = \int_{\alpha_1-\epsilon}^{\alpha_1+\epsilon} g(\omega)h(\omega)d\omega > 0$, and we have shown that, in the limit as $r_1 \to 1^-$, the integral (24) is positive. This means that poles that are close to the unit circle will not move outside the unit circle as a result of the regularization approach. ■

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